

	pose scientific questions and define scientific problems in investigative and practical units 3.3.1 • presentation of logical scientific argument in the synoptic essay in Unit 5, 3.5
C	Use appropriate methodology, including ICT, to answer scientific questions and solve scientific problems Observations ultimately lead to explanations in the form of hypotheses. In turn, these hypotheses lead to predictions that can be tested experimentally. Observations are one of the key links between the 'real world' and the abstract ideas of science. Once an experimental method has been validated, it becomes a protocol that is used by other scientists. ICT can be used to collect, record and analyse experimental data. <i>Examples in this specification include:</i> • choice of a suitable method to measure a variable, 3.3.1 • choice and use of a suitable statistical test, 3.6.3
D	Carry out experimental and investigative activities, including appropriate risk management, in a range of contexts Scientists perform a range of experimental skills that include manual and data handling skills (tabulation, graphical skills etc). Scientists should select and use equipment that is appropriate when making accurate measurements and should record these measurements methodically. Scientists carry out experimental work in such a way as to minimise the risk to themselves, to others and to the materials, including organisms, used. <i>Examples in this specification include:</i> • Many opportunities in the Investigative and practical skills units for AS and A2

E	Analyse and interpret data to provide evidence, recognising correlations and causal relationships Scientists look for patterns and trends in data as a first step in providing explanations of phenomena. The degree of uncertainty in any data will affect whether alternative explanations can be given for the data. Anomalous data are those measurements that fall outside the normal, or expected, range of measured values. Decisions on how to treat anomalous data should be made only after examination of the event. In searching for causal links between factors, scientists propose predictive theoretical models that can be tested experimentally. When experimental data confirm predictions from these theoretical models, scientists become confident that a causal relationship exists. <i>Examples in this specification include:</i> - analyse and interpret data associated with specific risk factors and the incidence of lung disease, 3.1.4 - analyse and interpret data relating to the distribution of organisms, recognising correlations and causal relationships, 3.4.1
F	Evaluate methodology, evidence and data, and resolve conflicting evidence The validity of new evidence, and the robustness of conclusions that stem from it, is constantly questioned by scientists. Experimental methods must be designed adequately to test predictions. Solutions to scientific problems are often developed when different research teams produce conflicting evidence. Such evidence is a stimulus for further scientific investigation, which involves refinements of experimental technique or development of new hypotheses <i>Examples in this specification include:</i> - evaluate methodology, evidence and data relating to the use of vaccines and monoclonal antibodies, 3.1.6 - evaluate evidence and data concerning issues relating to the conservation of species and habitats and consider conflicting evidence, 3.4.7
G	Appreciate the tentative nature of scientific knowledge Scientific explanations are those that are based on experimental evidence which is supported by the scientific community. Scientific knowledge changes when new evidence provides a better explanation of scientific observations. <i>Examples in this specification include:</i> - appreciate the tentative nature of any conclusions that can be drawn relating to the causes of variation, 3.2.1 - appreciate the tentative nature of conclusions that may be drawn from data relating to populations, 3.4.1
H	Communicate information and ideas in appropriate ways using appropriate terminology By sharing the findings of their research, scientists provide the scientific community with opportunities to replicate and further test their work, thus either confirming new explanations or refuting them. Scientific terminology avoids confusion amongst the scientific community, enabling better understanding and testing of scientific explanations. <i>Examples in this specification include:</i> Many opportunities through assessment of questions requiring extended prose for AS and A2

I	Consider applications and implications of science and appreciate their associated benefits and risks Scientific advances have greatly improved the quality of life for the majority of people. Developments in technology, medicine and materials continue to further these improvements at an increasing rate. Scientists can predict and report on some of the beneficial applications of their experimental findings. Scientists evaluate, and report on, the risks associated with the techniques they develop and the applications of their findings. <i>Examples in this specification include:</i> • interpret data relating to the effects of human activity on species diversity and be able to evaluate associated benefits and risks, 3.2.11 • balance the humanitarian aspects of recombinant DNA technology with opposition from environmentalists and anti-globalisation activists, 3.5.8
J	Consider ethical issues in the treatment of humans, other organisms and the environment Scientific research is funded by society, either through public funding or through private companies that obtain their income from commercial activities. Scientists have a duty to consider ethical issues associated with their findings. Individual scientists have ethical codes that are often based on humanistic, moral and religious beliefs. Scientists are self-regulating and contribute to decision-making about what investigations and methodologies should be permitted. <i>Examples in this specification include:</i> • discuss ethical issues involved in the selection of domesticated animals, 3.2.3 • evaluate evidence and data concerning issues relating to the conservation of species and habitats and consider conflicting evidence, 3.4.7
K	Appreciate the role of the scientific community in validating new knowledge and ensuring integrity The findings of scientists are subject to peer review before being accepted for publication in a reputable scientific journal. The interests of the organisations that fund scientific research can influence the direction of research. In some cases, the validity of those claims may also be influenced. <i>Examples in this specification include:</i> • explain the role of the scientific community in validating new knowledge about vaccines and monoclonal antibodies thus ensuring integrity, 3.1.6 • evaluate the ethical, moral and social issues associated with the use of recombinant technology in agriculture, in industry and in medicine, 3.5.8
L	Appreciate the ways in which society uses science to inform decision-making Scientific findings and technologies enable advances to be made that have potential benefit for humans. In practice, the scientific evidence available to decision-makers may be incomplete. Decision-makers are influenced in many ways, including by their prior beliefs, their vested interests, special interest groups, public opinion and the media, as well as by expert scientific evidence. <i>Examples in this specification include:</i> • discuss the ways in which society uses scientific knowledge relating to vaccines and monoclonal antibodies to inform decision-making, 3.1.6 • explain how conservation relies on science to inform decision-making, 3.4.7